

Learning Latex with Overleaf

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Introduction

This document is designed for absolute beginners learning Latex using Overleaf. Each lab introduces one concept, explains it clearly in simple language, and then presents examples using tables that show the **LaTeX code** and its **rendered output**.

Step 0: Registering on Overleaf

Before writing any Latex code, you must create an account on Overleaf. Overleaf is an online editor that allows you to write, compile, and preview Latex documents directly in your web browser without installing any software. This step is mandatory to follow all the next labs.

Action	Result
Open the Overleaf website : <code>https://www.overleaf.com</code>	Access to the Overleaf home page
Click <i>Sign Up</i>	Registration form is displayed
Register using email or Google	Overleaf account is created
Log in to your account	Access to the project dashboard
Click <i>New Project</i>	Ready to create a Latex document

Table 1: Overleaf registration process

Step 1: Minimal Latex Document

Every Latex document must follow a basic structure. This structure tells the compiler what type of document you are writing and where the actual content starts and ends. Without this structure, Latex cannot produce a PDF.

LaTeX Code	Output
<code>\documentclass{article}</code>	Defines the document type
<code>\begin{document} Hello \end{document}</code>	Hello

Table 2: Minimal document structure

Step 2: Paragraphs and Line Breaks

In $\text{\LaTeX}$, text formatting is automatic. A new paragraph is created by leaving an empty line in the source code. A manual line break can be forced using double backslashes.

LaTeX Code	Output
First paragraph.	First paragraph.
First.\\Second. Second.	First.

Table 3: Paragraphs and line breaks

Step 3: Title Creation

Titles in Latex are created using predefined commands. These commands store the title information, and the command `\maketitle` prints it in the document.

LaTeX Code	Output
<code>\title{My Document}</code>	Document title
<code>\author{Student}</code>	Author name
<code>\date{\today}</code>	Current date
<code>\maketitle</code>	Displays the title

Table 4: Title commands

Step 4: Sections and Subsections

Sections organize a document logically. Latex automatically numbers sections, subsections, and subsubsections, ensuring consistent structure throughout the document.

LaTeX Code	Output
<code>\section{Introduction}</code>	1 Introduction
<code>\subsection{Goal}</code>	1.1 Goal
<code>\subsubsection{Detail}</code>	1.1.1 Detail

Table 5: Document structure

Step 5: Text Formatting

Text formatting in Latex uses commands rather than buttons. This ensures consistent appearance and avoids manual formatting errors.

LaTeX Code	Output
<code>\textbf{Bold}</code>	Bold
<code>\textit{Italic}</code>	<i>Italic</i>
<code>\underline{Text}</code>	<u>Text</u>

Table 6: Text emphasis

Step 6: Inline Mathematics

Inline mathematics is used when mathematical expressions appear inside a sentence. The expression is written between dollar signs.

LaTeX Code	Output
<code>\$a^2 + b^2 = c^2\$</code>	$a^2 + b^2 = c^2$

Table 7: Inline math

Step 7: Displayed Mathematics

Displayed mathematics is used for important formulas that need to be centered and clearly visible on their own line.

LaTeX Code	Output
<code>\[x^2 + y^2 = z^2 \]</code>	$x^2 + y^2 = z^2$

Table 8: Displayed equations

Step 8: Fractions and Roots

Fractions and roots are fundamental in mathematics. Latex provides simple commands to typeset them correctly and professionally.

LaTeX Code	Output
<code>\frac{a}{b}</code>	$\frac{a}{b}$
<code>\sqrt{x}</code>	$\sqrt{x}$

Table 9: Math operators

Step 9: Superscripts and Subscripts

Superscripts and subscripts are used for powers, indices, and scientific notation. They are created using caret (^) and underscore (_) symbols.

LaTeX Code	Output
<code>x^2</code>	x^2
<code>x_i</code>	x_i

Table 10: Indices

Step 10: Lists

Lists help organize information clearly. Latex supports bullet lists and numbered lists using predefined environments.

LaTeX Code	Output
<code>\begin{itemize} \item A \item B \end{itemize}</code>	• A • B

Table 11: Itemized lists